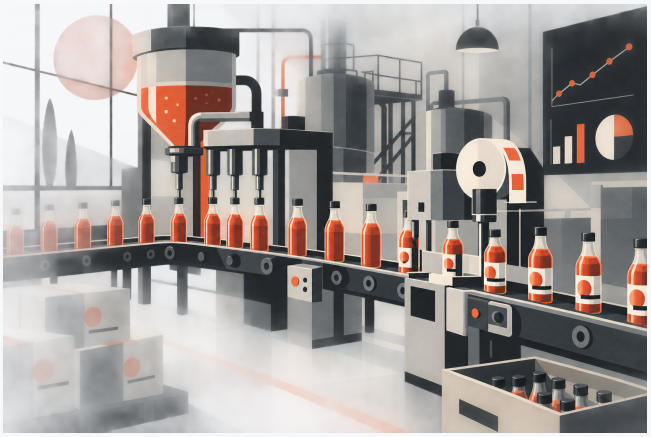
Computer-Based Data Processing

Lab 3 – Goal Seek and Optimisation



Objectives

In the previous labs, we learned how to transform and summarise data.

We will now use Calc to answer a new question:

How can we choose the best possible values subject to constraints?

We will work with a small company that produces several drinks.

By the end of this lab, you should be able to:

- build a simple model;
- use Goal Seek;
- define an objective, variables, and constraints;
- use Solver in LibreOffice Calc.

Open the file: `production_boissons.csv`

1 Building the model

Each row corresponds to a drink.

The file contains:

- the selling price;
- the unit cost;
- the machine time required;
- the amount of raw material required;
- the maximum demand.

Add the following columns:

Column	Calculation
`Quantity`	chosen value
`Unit_margin`	selling price − unit cost
`Revenue`	price × quantity
`Total_cost`	cost × quantity
`Profit`	margin × quantity
`Time_used`	machine time × quantity
`Material_used`	material × quantity

Initially, enter `0` in every cell of the `Quantity` column.

Write the formulas for the first drink, then copy them down to the other rows.

Next, add a **TOTAL** row to calculate:

- total revenue;
- total cost;
- total profit;
- machine time used;
- raw material used.

For example:

```
=SUM(I2:I5)
```

Change a few quantities manually and observe the effect on the results.

What to look for: the cells containing the quantities are the inputs to your model. When you change them, all values calculated from these quantities are updated automatically.

2 Goal Seek

Suppose the company now produces only Cola.

It wants to achieve a profit of exactly:

350 €

Reset all quantities to zero, then open:

Tools → Goal Seek

Use:

- total profit as the **formula cell**;
- `350` as the **target value**;
- the quantity of Cola as the **variable cell**.

Run Goal Seek.

Check the suggested quantity and the resulting profit.

Goal Seek finds an **input value** from a **desired result**.

However, it can only change one cell at a time.

What if we want to choose the quantities of all four products simultaneously?

3 Optimising with Solver

The company has the following resources:

Resource	Available
Machine time	5000 min
Raw material	500 L
Production budget	800 €

It now wants to determine how much of each drink to produce in order to:

maximise total profit.

The quantities produced are the **decision variables**.

Open:

Tools → Solver

Configure:

- **target cell**: total profit;
- **optimisation**: maximum;
- **variable cells**: the four quantities.

Add the following constraints:

$$\text{Time used} \leq 5000$$

$$\text{Material used} \leq 500$$

$$\text{Total cost} \leq 800$$

For each product:

$$0 \leq \text{Quantity} \leq \text{Maximum demand}$$

Run Solver.

Examine the proposed solution:

- which drinks are produced?
- which products reach their maximum demand?
- which resources are fully used?
- what is the maximum profit?

4 Producing whole-number quantities

Solver may suggest quantities with decimal values.

However, we cannot produce a fraction of a bottle.

Add a constraint requiring the quantities to be **integers**, then run Solver again.

Compare this solution with the previous one.

5 Changing the problem

Now change some constraints and run Solver again.

Scenario 1

The available machine time decreases from:

$$5000 \quad \text{to} \quad 3500 \text{ minutes.}$$

Observe how production and profit change.

Scenario 2

Restore the available machine time to 5000 minutes.

The available amount of raw material increases from:

$$500 \quad \text{to} \quad 600 \text{ L.}$$

Does this additional resource actually improve the solution?

Scenario 3

The maximum demand for Root Beer increases from:

$$200 \quad \text{to} \quad 400.$$

Run the optimisation again and observe how the production strategy changes.

Further practice

Choose one of the following:

- require a minimum production of 50 units for each product;
- add a fifth drink to the model;
- replace the objective *maximise profit* with *maximise revenue*.

Review

In this lab, we used Calc not only to calculate or summarise data, but also to **make a decision**.

Lab	Task	Main tool
Lab 1	Transform	Formulas
Lab 2	Summarise	Pivot tables
Lab 3	Optimise	Goal Seek / Solver